

Development of a Mathematical Model for the Simulation of Ex Vivo Ocular Tissue Clearing

Proof of “Changes in pore pressure differential” Equation

Proof. To model changes in pore pressure, the matrix compressibility of the layer must be considered. The Continuity Equation of Mass and Standard Hydraulic Diffusivity Formula were used to structure the equation,

$$D = \frac{K}{S_s}, S_s = \rho g(\varepsilon C_f + C_s)$$

By convention, fluid compressibility and matrix compressibility can both be equated in terms of fluid velocity and pressure:

$$C_f = -\frac{1}{q} \frac{\partial q}{\partial p}, C_s = \frac{1}{\varepsilon} \frac{\partial \varepsilon}{\partial p}$$

Based on the principles of partial derivatives, the compressibilities may also be expressed as

$$\frac{1}{p} \frac{\partial p}{\partial t} = C_f \frac{\partial p}{\partial t}, \frac{\partial \varepsilon}{\partial t} = \varepsilon C_s \frac{\partial p}{\partial t}$$

Rewriting the differential equation,

$$\frac{\partial p \varepsilon}{\partial t} = \varepsilon \frac{\partial p}{\partial t} + p \frac{\partial \varepsilon}{\partial t}$$

dividing by p on both sides then substituting it into the Continuity Equation,

$$\varepsilon \frac{1}{p} \frac{\partial p}{\partial t} + \frac{\partial \varepsilon}{\partial t} + \nabla \cdot q = 0.$$

The redefined compressibility terms are substituted, yielding

$$\varepsilon(C_f \frac{\partial p}{\partial t}) + \varepsilon C_s \frac{\partial p}{\partial t} + \nabla \cdot q = 0,$$

which simplifies to

$$\varepsilon(C_f + C_s) \frac{\partial p}{\partial t} + \nabla \cdot q = 0.$$

Substituting Darcy's Law in place of the divergence of fluid velocity and adding the effect of CUBIC diffusion, the derived equation in terms of pore pressure change becomes:

$$\boxed{\frac{\partial p}{\partial t} = \frac{k}{\mu \varepsilon (C_f + C_s)} \nabla^2 p + \alpha \frac{\partial C}{\partial t}}.$$